

BINLEVEL · DESIGN DECISION RECORD

BinLevel recommendations accepted

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DOCUMENT CONTROL

Document ID	BNL-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/binlevel		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	01 and 02 decided by Amish on 2026-10-02 (recommendations approved, BNL-DEC-001)
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all recommendations and the changes made in the repo

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below that carried a recommendation is "Decided by Amish, 2026-09-25: go with recommendation". Items without a recommendation (O1, O2) were decided on 2026-10-02, when Amish approved the recommendations later written for them ("i approve your recommendations for all 555 open decisions."); see Table 2 and BNL-DEC-001.

■ Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." BinLevel had ten items adopted for TRL 3 under his earlier instruction and open for his review (BNL-DDR-001, D1 to D10), four new items raised by the sizing note BNL-CAL-001 v0.1 with a recommendation each, one suggestion on the reporting interval, and two items with no recommendation (O1, O2). Where a recommendation offered several options, the recommended option is the decision. TRL 4 remains on hold by Amish's instruction, and the repo stays at TRL 3.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1 to D10	TRL 2 review items (BNL-DDR-001): communal containers first, ultrasonic plus ToF, C cell, LoRaWAN on RAK3172, FieldNode and TwinKit reuse, 80 % and 1 h defaults, heat alert, lid mounting, stock IP67 enclosure, no budget change	As recommended	Status wording in BNL-DDR-001 (v0.2), BNL-PRB-001, BNL-PRC-001 and README; no design change
N1	R16 mass: (a) 2 mm aluminium bracket, (b) relax R16 to 450 g, (c) perforated stainless	(a) 2 mm 5052-class aluminium bracket	<code>cad/src/model.py</code> plate 1.5 mm stainless to 2.0 mm aluminium; STEP and STL re-exported; BOM item 2 renamed and respecified (price unchanged at \$5.00); drawing BNL-DWG-001 to Rev P2; BNL-CAL-001 v0.2: unit mass 435 g to 334 g, envelope below the lid 61.0 to 61.5 mm, shock load 85 to 66 N; R16 from not met to met
N2	R6 upper limit	Raise from 60 to 70 °C; name the RAK3172-T (-40 to 85 °C) for cold sites	BNL-REQ-001 R6 target; BNL-CAL-001 speed-of-sound range now -20 to 70 °C (uncompensated error up to +8.2 %, was +6.6 %); R6 stays at risk (3 K margin, transducer rating unconfirmed)
N3	R9 rate-of-rise rule	Count the 15 K rise in 15 min only when the unit is already above 50 °C; keep the 70 °C absolute alert	BNL-REQ-001 R9 target; BNL-PRC-001 firmware rule; BNL-CAL-001 new checks H3 and H4 (unit about 43 °C under cloud, below the gate; worst rise above the gate 10.9 K); R9 from at risk to met on paper
N4	11-byte payload and 5 min temperature reads	Keep both, as applied at TRL 3	Status wording only
N5	Reporting interval stretch (suggestion in BNL-CAL-001 v0.1)	Stretch the routine interval to 2 h at SF12 only; SF11 stays hourly (21.4 s a day, inside TTN's 30 s)	BNL-REQ-001 R3, BNL-PRC-001, BNL-CAL-001 section E, concept media key figures; worst-case battery life unchanged at 16.3 years
N6	Budget	No change was recommended	<code>budget_usd</code> stays \$60; parts total stays \$55.00

Table 2. Items left open on 2026-09-25, decided on 2026-10-02.

#	Item	Status
O1	Steel containers: (a) external lid antenna variant (about \$8, total \$63.00, over budget) or (b) exclude steel containers from the first pilot. No recommendation was made.	Decided by Amish, 2026-10-02 (recommendation approved): (b), steel containers left out of the first pilot, which uses plastic containers only; the external lid antenna is revisited once the link loss inside a steel container has been measured (BNL-DEC-001)
O2	First partner and region for co-design and a pilot; also sets the radio plan (EU868, US915 or IN865). No preference stated.	Decided by Amish, 2026-10-02 (recommendation approved): first partner to approach is a municipal waste service or its contractor using EN 840 plastic communal containers in a city with public LoRaWAN coverage, by default a European city on EU868; washing practice asked before committing (BNL-DEC-001)

Table 3. Decided but on hold or outside this repo.

Item	Status
TRL 4 work named in the review note (bench build, lab test report, ranging on real waste, current and cold-start measurements, link loss in HDPE and steel containers, lid temperature in sun, shake and drop)	Decided as the next step, on hold: TRL 4 is on hold by Amish's instruction
FieldNode's 20-byte payload exceeds the 11-byte US915 DR0 limit found here, and FieldNode stretches its interval from SF10	Cross-repo action for FieldNode, listed in docs/REVIEW.md ; FieldNode not edited

■ Consequences

- Requirement status (BNL-CAL-001 v0.2): not met 2 (R5 IP69K; R10 in steel containers), at risk 2 (R2, R6), not verifiable at TRL 3 2 (R7, R12), met 10 (R1, R3, R4, R9, R14, R16 by calculation; R8, R11, R13, R15 by design). Before this record: not met 3, at risk 3, met 8.
- Controlled documents bumped: BNL-PRB-001 v0.4, BNL-PRC-001 v0.4, BNL-REQ-001 v0.4, BNL-CAL-001 v0.2, BNL-DDR-001 v0.2; drawing BNL-DWG-001 Rev P2.
- The stainless bolts now pass through an aluminium plate. Galvanic isolation (anodized plate or insulating washers) is a review suggestion, not a decision.
- `trl: 3` and `trl_target: 3` are unchanged.